The Singular Value Decomposition

Linear Algebra · Lecture 13

Last time (Lecture 12): if $a'\beta$ is estimable, $a'\beta^0$ is its BLUE; t intervals and tests, the F test of a testable hypothesis, and the one-way analysis-of-variance table. Throughout, I and $e_j = \mathrm{col}_j(I)$ are of the size required; $\lambda_1(M) \geqslant \cdots \geqslant \lambda_n(M)$ are the eigenvalues of a symmetric $M \in \mathbb{R}^{n\times n}$; $\mathrm{diag}(D, 0)$ is the block matrix of the size required by the product, with $D \in \mathbb{R}^{r\times r}$ in its top left block and zeros elsewhere (blocks with no rows or no columns are absent).

From earlier lectures (restated).

- Lecture 1, Theorem 3(a), Definitions 7, 9, Theorems 8, 10(a), Proposition 13(a),(b); Lecture 2, Definition 10; Lecture 7, Definition 1: matrix products are associative and distributive; $(A')' = A$, $(A+B)' = A' + B'$, $(cA)' = cA'$, $(AB)' = B'A'$; A is **symmetric** if $A' = A$, and $B'B$, BB' are symmetric; $\mathrm{tr}\, A = \sum_i a_{ii}$, $\mathrm{tr}(A+B) = \mathrm{tr}\, A + \mathrm{tr}\, B$, $\mathrm{tr}(cA) = c\,\mathrm{tr}\, A$; for $D = \mathrm{diag}(d_1, \ldots, d_n)$: $\mathrm{col}_j(AD) = d_j \mathrm{col}_j(A)$, $D\,\mathrm{diag}(f_1, \ldots, f_n) = \mathrm{diag}(d_1 f_1, \ldots, d_n f_n)$, and $D^{-1} = \mathrm{diag}(1/d_1, \ldots, 1/d_n)$ if all $d_i \neq 0$; if $Ax = \lambda x$, $\lambda \in \mathbb{R}$, $x \neq 0$, then x is an **eigenvector** of A belonging to λ.
- Lecture 1, Theorem 4(a),(b),(d): for $A = [a_1 \ \cdots \ a_n] \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times p}$: $\mathrm{col}_j(AB) = A\,\mathrm{col}_j(B)$; $Ax = x_1 a_1 + \cdots + x_n a_n$, $Ae_j = a_j$; $(AB)_{ij} = \mathrm{row}_i(A)\,\mathrm{col}_j(B)$ and $AB = a_1\,\mathrm{row}_1(B) + \cdots + a_n\,\mathrm{row}_n(B)$.
- Lecture 1, Theorem 10(b),(c): $\mathrm{tr}(AB) = \mathrm{tr}(BA)$ ($A \in \mathbb{R}^{m\times n}$, $B \in \mathbb{R}^{n\times m}$); $\mathrm{tr}(A'A) = \sum_i \sum_j a_{ij}^2$, so $\mathrm{tr}(A'A) = 0 \Rightarrow A = 0$.
- Lecture 2, Theorem 11(b): $(AB)^{-1} = B^{-1}A^{-1}$ for invertible $A, B \in \mathbb{R}^{n\times n}$.
- Lecture 4, Definitions 7, 10, Theorems 11, 12(a),(d), 15(d), 16(a),(b): for $A \in \mathbb{R}^{m\times n}$, $\mathcal{C}(A) = \{Ax : x \in \mathbb{R}^n\}$, $\mathcal{N}(A) = \{x : Ax = 0\}$, $\mathrm{rank}\, A = \dim \mathcal{C}(A)$; $\mathrm{rank}\, A + \dim \mathcal{N}(A) = n$; $\mathrm{rank}\, A' = \mathrm{rank}\, A \leqslant \min\{m, n\}$; if $m = n$, A is invertible $\iff$ $\mathrm{rank}\, A = n$; $\mathrm{rank}(PAQ) = \mathrm{rank}\, A$ for invertible P, Q; $\mathcal{N}(A'A) = \mathcal{N}(A)$ and $\mathrm{rank}(A'A) = \mathrm{rank}\, A = \mathrm{rank}(AA')$.
- Lecture 5, Theorem 2(b),(c): block matrices are transposed block by block (block (j, i) of A' is A'_{ij}) and multiplied block by block (when the blocks fit).
- Lecture 6, Definitions 1, 4, 8, Theorems 5(c), 7(b), 9: $\|x\| = \sqrt{x'x}$, which is 0 only for $x = 0$; $\|cx\| = |c|\,\|x\|$ ($c \in \mathbb{R}$); a **unit vector** has length 1, and $x/\|x\|$ is one for $x \neq 0$. For a subspace $W \subseteq \mathbb{R}^n$, $W^\perp = \{y : y'w = 0 \text{ for all } w \in W\}$ and $\dim W + \dim W^\perp = n$. Nonzero pairwise orthogonal vectors are linearly independent. $Q \in \mathbb{R}^{n\times p}$ has **orthonormal** columns if $Q'Q = I_p$; then $\|Qx\| = \|x\|$; if moreover $p = n$ (Q is **orthogonal**), $Q^{-1} = Q'$ and $QQ' = I_n$.
- Lecture 6, Corollary 11(a),(c): a subspace $W \neq \{0\}$ of $\mathbb{R}^n$ has an orthonormal basis. If $Q'Q = I_p$ and $p < n$, then $[Q\ T]$ is orthogonal for some $T \in \mathbb{R}^{n\times(n-p)}$, and $QQ' + TT' = I_n$ for every such T.
- Lecture 7, Definition 5, Theorem 14(a): $A \in \mathbb{R}^{n\times n}$ has n eigenvalues $\lambda_1, \ldots, \lambda_n$, listed with multiplicity; $a_A(c) = \#\{i : \lambda_i = c\}$. If P is invertible and $P^{-1}AP = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$, then $\lambda_1, \ldots, \lambda_n$ are the eigenvalues of A.
- Lecture 8, Theorems 8, 9(a),(b): let $A \in \mathbb{R}^{n\times n}$ be symmetric. Eigenvectors for different eigenvalues are orthogonal; $\mathbb{R}^n$ has an orthonormal basis of eigenvectors of A. If $q_1, \ldots, q_n$ are orthonormal, $Aq_i = \lambda_i q_i$ and $Q = [q_1 \ \cdots \ q_n]$: Q is orthogonal, $Q'AQ = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$, the λ_i are the eigenvalues of A, $\mathrm{rank}\, A = \#\{i : \lambda_i \neq 0\}$, and $\dim \mathcal{N}(A - cI) = a_A(c)$ ($c \in \mathbb{R}$).
- Lecture 8, Definition 13, Theorems 14(a), 15(b): a symmetric A is **positive definite** (PD) if $x'Ax > 0$ for all $x \neq 0$, **positive semidefinite** (PSD) if $x'Ax \geqslant 0$ for all x; A is PD $\iff$ every $\lambda_i > 0$, PSD $\iff$ every $\lambda_i \geqslant 0$. For $A \in \mathbb{R}^{n\times p}$, $A'A$ is PSD, and PD $\iff$ $\mathrm{rank}\, A = p$.
- Lecture 9, Theorem 13(a): for a symmetric A, orthonormal $q_1, \ldots, q_n$, $Aq_i = \lambda_i q_i$, can be numbered so that $\lambda_1 \geqslant \cdots \geqslant \lambda_n$; then $Q = [q_1 \ \cdots \ q_n]$ is orthogonal, $Q'AQ = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$, the λ_i are the eigenvalues of A, and $\lambda_n \leqslant x'Ax/x'x \leqslant \lambda_1$ for $x \neq 0$, with equality at $x = q_n$ and $x = q_1$.
- Lecture 1, Definition 14; Lecture 9, Definition 9, Proposition 10(a): for a data matrix $X \in \mathbb{R}^{n\times p}$ ($n \geqslant 2$), $C_n = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}'$ ($\mathbf{1} = (1, \ldots, 1)' \in \mathbb{R}^n$) and $S = \frac{1}{n-1}X'C_nX = \frac{1}{n-1}(C_nX)'(C_nX)$.
- Lecture 9, Definition 15, Theorem 16(a),(d), Proposition 17: let $S \neq 0$, $S = V\Lambda V'$, $V = [v_1 \ \cdots \ v_p]$ orthogonal, $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_p)$, $\lambda_1 \geqslant \cdots \geqslant \lambda_p$, the score vectors $z_m = C_nXv_m$, and $r =$

$\operatorname{rank}(C_nX)$. Then $\lambda_p \geqslant 0$, $\lambda_m > 0 \iff m \leqslant r$; $z_m'z_l = (n-1)\lambda_m$ if $l = m$, 0 if $l \neq m$; $\mathrm{PVE}_m = \|z_m\|^2 / \operatorname{tr}\Big((C_nX)'(C_nX)\Big) = \lambda_m/(\lambda_1 + \cdots + \lambda_p)$.

- Lecture 9, Examples 11, 18: $n = 5$, $p = 4$, $X = \begin{pmatrix} 7&3&7&3\\ 9&9&5&7\\ 5&7&9&9\\ 3&5&1&1\\ 1&1&3&5 \end{pmatrix}$, $C_5X = \begin{pmatrix} 2&-2&2&-2\\ 4&4&0&2\\ 0&2&4&4\\ -2&0&-4&-4\\ -4&-4&-2&0 \end{pmatrix}$, $S = \begin{pmatrix} 10&7&5&3\\ 7&10&3&5\\ 5&3&10&7\\ 3&5&7&10 \end{pmatrix}$; $\lambda_m = 25, 9, 5, 1$, $v_m = \frac{1}{2}h_m$ with $h_1 = (1,1,1,1)'$, $h_2 = (1,1,-1,-1)'$, $h_3 = (1,-1,1,-1)'$, $h_4 = (1,-1,-1,1)'$; $z_1 = (0,5,5,-5,-5)'$, $z_2 = (0,3,-3,3,-3)'$, $z_3 = (4,-1,-1,-1,-1)'$, $z_4 = (0,1,-1,-1,1)'$.
- Lecture 11, Definition 1: in the linear model $y = X\beta + \varepsilon$, $X'X\beta = X'y$ are the **normal equations**.

1. The singular value decomposition

Theorem 1. Let $A \in \mathbb{R}^{m\times n}$, $\operatorname{rank} A = r \geqslant 1$.

(a) Let $Q = [Q_1\ Q_2] \in \mathbb{R}^{n\times n}$ be orthogonal, $Q_1 \in \mathbb{R}^{n\times r}$, and $D \in \mathbb{R}^{r\times r}$ invertible and diagonal, with $Q'A'AQ = \operatorname{diag}(D^2, 0)$. Let $P_1 = AQ_1D^{-1} \in \mathbb{R}^{m\times r}$ and $P = [P_1\ P_2]$, where $P_2 \in \mathbb{R}^{m\times(m-r)}$ and $P_1'P_2 = 0$. Then $P'AQ = \operatorname{diag}(D, 0)$.

(b) Such Q, D exist with $D = \operatorname{diag}(\sigma_1, \ldots, \sigma_r)$, $\sigma_1 \geqslant \cdots \geqslant \sigma_r > 0$: $\sigma_i^2 = \lambda_i(A'A)$ $(i \leqslant r)$, Q as in Lecture 9, Theorem 13, for $A'A$. Then $P_1'P_1 = I_r$, P_2 can be chosen so that P is orthogonal, and, with the columns p_i of P and q_i of Q,

$$A = P\operatorname{diag}(D,0)Q' = P_1DQ_1' = \sigma_1p_1q_1' + \cdots + \sigma_rp_rq_r'$$

Proof.

(a) The blocks of $Q'A'AQ$ (Lecture 5, Theorem 2(b),(c)):

$$\begin{pmatrix} Q_1'A'AQ_1 & Q_1'A'AQ_2 \\ Q_2'A'AQ_1 & Q_2'A'AQ_2 \end{pmatrix} = \begin{pmatrix} D^2 & 0 \\ 0 & 0 \end{pmatrix}$$

$$(AQ_2)'(AQ_2) = Q_2'A'AQ_2 = 0$$

$$\implies \operatorname{tr}\Big((AQ_2)'(AQ_2)\Big) = 0 \implies AQ_2 = 0 \qquad \text{(Lecture 1, Theorem 10(c))}$$

(a) $AQ_1 = P_1D$; $P_1' = D^{-1}Q_1'A'$ (D^{-1} is diagonal); Lecture 5, Theorem 2(b),(c):

$$P'AQ = \begin{pmatrix} P_1'AQ_1 & P_1'AQ_2 \\ P_2'AQ_1 & P_2'AQ_2 \end{pmatrix} = \begin{pmatrix} D^{-1}Q_1'A'AQ_1 & 0 \\ P_2'P_1D & 0 \end{pmatrix} = \begin{pmatrix} D^{-1}D^2 & 0 \\ (P_1'P_2)'D & 0 \end{pmatrix} = \begin{pmatrix} D & 0 \\ 0 & 0 \end{pmatrix}$$

(b) D and Q: $A'A$ is symmetric; $\lambda_i = \lambda_i(A'A)$:

$$A'A \text{ is PSD} \implies \lambda_i \geqslant 0 \qquad \text{(Lecture 8, Theorems 15(b), 14(a))}$$

$$\operatorname{rank}(A'A) = \operatorname{rank} A = r \qquad \text{(Lecture 4, Theorem 16(b))}$$

$$\implies \text{exactly } r \text{ of the } \lambda_i \text{ are nonzero} \qquad \text{(Lecture 8, Theorem 9(b))}$$

$$\lambda_1 \geqslant \cdots \geqslant \lambda_r > 0 = \lambda_{r+1} = \cdots = \lambda_n$$

$$Q'A'AQ = \operatorname{diag}(\lambda_1, \ldots, \lambda_n) = \operatorname{diag}(D^2, 0) \qquad \text{(Lecture 9, Theorem 13)}$$

(b) P is orthogonal:

$$P_1'P_1 = D^{-1}Q_1'A'AQ_1D^{-1} = D^{-1}D^2D^{-1} = I_r \qquad \text{((a))}$$

$$r < m\text{: } [P_1\ T] \text{ is orthogonal for some } T \qquad \text{(Lecture 6, Corollary 11(c))}$$

$$P_2 = T\text{: } P'P = I_m \implies P_1'P_2 = 0 \qquad \text{(Lecture 5, Theorem 2(c))}$$

$$r = m\text{: } P = P_1,\ P'P = I_m$$

(b) The three forms: $PP' = I_m$, $QQ' = I_n$ (Lecture 6, Theorem 9):

$$\begin{aligned} A = P(P'AQ)Q' &= P\,\mathrm{diag}(D,0)Q' && ((a)) \\ &= [P_1 D \;\; 0]\begin{pmatrix} Q_1' \\ Q_2' \end{pmatrix} = P_1 D Q_1' && \text{(Lecture 5, Theorem 2(b),(c))} \\ &= \sigma_1 p_1 q_1' + \cdots + \sigma_r p_r q_r' && \text{(Lecture 1, Theorem 4(a),(d): } \mathrm{col}_i(P_1 D) = \sigma_i p_i) \end{aligned}$$

□

Definition 2. A factorization $A = P\,\mathrm{diag}(D,0)Q'$ with orthogonal $P \in \mathbb{R}^{m\times m}$, $Q \in \mathbb{R}^{n\times n}$ and $D = \mathrm{diag}(\sigma_1, \ldots, \sigma_r)$, $\sigma_1 \geqslant \cdots \geqslant \sigma_r > 0$, is a **singular value decomposition** (SVD) of $A \in \mathbb{R}^{m\times n}$; $A = P_1 D Q_1'$ is the **compact SVD**. $\sigma_1, \ldots, \sigma_r$ are the **singular values** of A (by Theorem 3(a), $r = \mathrm{rank}\, A$ and they depend only on A), and the columns of P and Q are **singular vectors** of A; $\sigma_k = 0$ for $r < k \leqslant \min\{m,n\}$, and $\sigma_1 = 0$ for $A = 0$. The **Frobenius norm** of A is $\|A\|_F = \big(\mathrm{tr}(A'A)\big)^{1/2} = \big(\sum_i \sum_j a_{ij}^2\big)^{1/2}$ (Lecture 1, Theorem 10(c)).

Theorem 3. Let $A \in \mathbb{R}^{m\times n}$, $P \in \mathbb{R}^{m\times m}$ and $Q \in \mathbb{R}^{n\times n}$ orthogonal, $D = \mathrm{diag}(\sigma_1, \ldots, \sigma_r)$, $\sigma_1 \geqslant \cdots \geqslant \sigma_r > 0$, $P'AQ = \mathrm{diag}(D,0)$; $P = [P_1 \; P_2]$, $Q = [Q_1 \; Q_2]$, P_1 with the columns $p_1, \ldots, p_r$, Q_1 with $q_1, \ldots, q_r$.

(a) $r = \mathrm{rank}\, A$; $\sigma_1^2, \ldots, \sigma_r^2$ are the nonzero eigenvalues of $A'A$ and of AA', listed with multiplicity.

(b) $Q'A'AQ = \mathrm{diag}(D^2,0)$, $P'AA'P = \mathrm{diag}(D^2,0)$: the columns of Q (of P) are orthonormal eigenvectors of $A'A$ (of AA'); q_i, p_i belong to σ_i^2 $(i \leqslant r)$, the other columns to 0.

(c) $P_1 = AQ_1D^{-1}$, $Q_1 = A'P_1D^{-1}$: $Aq_i = \sigma_i p_i$, $A'p_i = \sigma_i q_i$ $(i \leqslant r)$.

(d) $\|A\|_F^2 = \sigma_1^2 + \cdots + \sigma_r^2$.

(e) If $A \neq 0$ is PSD, its singular values are its nonzero eigenvalues. For orthogonal $U \in \mathbb{R}^{m\times m}$, $V \in \mathbb{R}^{n\times n}$, the matrices UAV and A' have the singular values of A.

Proof.

(a), (b) $PP' = I_m$, $QQ' = I_n$, $Q^{-1} = Q'$ (Lecture 6, Theorem 9); the nonzero columns $\sigma_1 e_1, \ldots, \sigma_r e_r$ of $P'AQ$ are independent (Lecture 6, Theorem 7(b)) and span $\mathcal{C}(P'AQ)$:

$$\begin{aligned} \mathrm{rank}\, A = \mathrm{rank}(P'AQ) = r && \text{(Lecture 4, Theorem 15(d))} \\ Q'A'AQ = Q'A'PP'AQ = (P'AQ)'(P'AQ) = \mathrm{diag}(D^2,0) && \text{(Lecture 5, Theorem 2)} \\ P'AA'P = (P'AQ)(P'AQ)' = \mathrm{diag}(D^2,0) && \\ Q^{-1}(A'A)Q = \mathrm{diag}(\sigma_1^2, \ldots, \sigma_r^2, 0, \ldots, 0) && \text{(Lecture 7, Theorem 14(a))} \\ A'AQ = Q\,\mathrm{diag}(D^2,0)\colon\; A'Aq_i = \sigma_i^2 q_i \;(i \leqslant r),\; A'Aq_i = 0 \;(i > r) && \end{aligned}$$

So the σ_i^2 and 0 are the eigenvalues of $A'A$; the columns by Lecture 1, Theorem 4(a). Likewise for AA' with P.

(c) $PP' = P_1P_1' + P_2P_2' = I_m$, $QQ' = Q_1Q_1' + Q_2Q_2' = I_n$; $P_1'AQ_1 = D$, $P_2'AQ_1 = 0$, $P_1'AQ_2 = 0$ are blocks of $P'AQ$, and $Q_1'A'P_1 = (P_1'AQ_1)' = D$, $Q_2'A'P_1 = (P_1'AQ_2)' = 0$:

$$\begin{aligned} P_1 &= P_1(P_1'AQ_1)D^{-1} = (I_m - P_2P_2')AQ_1D^{-1} = AQ_1D^{-1} - P_2(P_2'AQ_1)D^{-1} = AQ_1D^{-1} \\ Q_1 &= Q_1(Q_1'A'P_1)D^{-1} = (I_n - Q_2Q_2')A'P_1D^{-1} = A'P_1D^{-1} \end{aligned}$$

The columns (Lecture 1, Theorem 4(a)): $Aq_i = \sigma_i p_i$, $A'p_i = \sigma_i q_i$.

(d) Lecture 1, Theorem 10(b),(c), and $QQ' = I_n$, $PP' = I_m$:

$$\|A\|_F^2 = \mathrm{tr}(A'A) = \mathrm{tr}(A'AQQ') = \mathrm{tr}(Q'A'PP'AQ) = \|P'AQ\|_F^2 = \sigma_1^2 + \cdots + \sigma_r^2$$

(e) $A \neq 0$ PSD; H, λ_i: the Q, λ_i of Lecture 9, Theorem 13, for A; $r = \operatorname{rank} A$:

$\lambda_i \geqslant 0$, exactly r nonzero $\implies \lambda_1 \geqslant \cdots \geqslant \lambda_r > 0 = \lambda_{r+1} = \ldots$ (Lecture 8, Theorems 14(a), 9(b))
$H'AH = \operatorname{diag}(\lambda_1, \ldots, \lambda_n) = \operatorname{diag}(D, 0)$, $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_r)$
$\implies A = H \operatorname{diag}(D, 0) H'$ is an SVD (Definition 2) (Lecture 6, Theorem 9)
$\implies$ the singular values of A are $\lambda_1, \ldots, \lambda_r$ ((a))

(e) UAV and A', with the P, Q of the theorem:

$(UP)'(UP) = P'U'UP = I_m$, $(V'Q)'(V'Q) = Q'VV'Q = I_n$ (Lecture 6, Theorem 9)
$(UP)'(UAV)(V'Q) = P'AQ = \operatorname{diag}(D, 0)$: an SVD of UAV, same D
$Q'A'P = (P'AQ)' = \operatorname{diag}(D, 0)$ $(n \times m)$: an SVD of A', same D (Lecture 5, Theorem 2(b))
$\implies UAV$ and A' have the singular values $\sigma_1, \ldots, \sigma_r$ of A ((a))

□

Example 4. Find an SVD and the compact SVD of

$$A = \begin{pmatrix} 2 & 0 & 1 & 1 \\ 0 & 2 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$

Solution.
Columns $a_1, \ldots, a_4$ of A; $\operatorname{rank} A = \dim \mathcal{C}(A)$ (Lecture 4, Definition 10):

$a_3 = a_4 = \frac{1}{2}(a_1 + a_2)$, a_1, a_2 independent
$\implies r = \operatorname{rank} A = 2$
$A(0, 0, 1, -1)' = a_3 - a_4 = 0$
$A(1, 1, -1, -1)' = a_1 + a_2 - a_3 - a_4 = 0$

$A'A$ and its eigenvalues:

$$A'A = \begin{pmatrix} 5 & 1 & 3 & 3 \\ 1 & 5 & 3 & 3 \\ 3 & 3 & 3 & 3 \\ 3 & 3 & 3 & 3 \end{pmatrix}: \text{ every row sum is } 12$$

$A'A(1, 1, 1, 1)' = 12(1, 1, 1, 1)'$
$A'A(1, -1, 0, 0)' = 4(1, -1, 0, 0)'$
$\dim \mathcal{N}(A'A) = \dim \mathcal{N}(A) = 4 - 2 = 2$ (Lecture 4, Theorems 16(a), 11)
$a_{A'A}(c) = \dim \mathcal{N}(A'A - cI)$ (Lecture 8, Theorem 9(b))
$a_{A'A}(12) \geqslant 1$, $a_{A'A}(4) \geqslant 1$, $a_{A'A}(0) = 2$
the $a_{A'A}(c)$ add up to 4 $\implies$ eigenvalues $12, 4, 0, 0$ (Lecture 7, Definition 5)

Check: $\operatorname{tr}(A'A) = 5 + 5 + 3 + 3 = 16 = 12 + 4$ (Theorem 3(d)).
Theorem 1(b). By Lecture 8, Theorem 8, and $(1, 1, -1, -1)(0, 0, 1, -1)' = 0$, the four vectors are pairwise orthogonal; normalize them:

$q_1 = \frac{1}{2}(1, 1, 1, 1)'$, $q_2 = \frac{1}{\sqrt{2}}(1, -1, 0, 0)'$, $q_3 = \frac{1}{2}(1, 1, -1, -1)'$, $q_4 = \frac{1}{\sqrt{2}}(0, 0, 1, -1)'$
$D = \operatorname{diag}(\sqrt{12}, 2) = \operatorname{diag}(2\sqrt{3}, 2)$
$Aq_1 = (2, 2, 2)'$, $Aq_2 = \frac{1}{\sqrt{2}}(2, -2, 0)' \implies p_1 = \frac{1}{2\sqrt{3}} Aq_1 = \frac{1}{\sqrt{3}}(1, 1, 1)'$, $p_2 = \frac{1}{2} Aq_2 = \frac{1}{\sqrt{2}}(1, -1, 0)'$
$P_1'y = 0$: $y_1 + y_2 + y_3 = 0$, $y_1 - y_2 = 0 \implies y = t(1, 1, -2)'$; $p_3 = \frac{1}{\sqrt{6}}(1, 1, -2)'$

P is orthogonal; $P_1'P_1 = I_2$ (Theorem 1(b)):

$P = [P_1\ p_3]$: $P_1'p_3 = 0$, $p_3'p_3 = 1 \implies P'P = I_3$ (Lecture 5, Theorem 2(b),(c))
$P'AQ = \operatorname{diag}(D, 0)$ (Theorem 1(a))

So $A = P(P'AQ)Q' = P\,\mathrm{diag}(D,0)Q'$ ($PP' = I_3$, $QQ' = I_4$; Lecture 6, Theorem 9) with

$$P = \begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & -\frac{2}{\sqrt{6}} \end{pmatrix}, \qquad \mathrm{diag}(D,0) = \begin{pmatrix} 2\sqrt{3} & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad Q = \begin{pmatrix} \frac{1}{2} & \frac{1}{\sqrt{2}} & \frac{1}{2} & 0 \\ \frac{1}{2} & -\frac{1}{\sqrt{2}} & \frac{1}{2} & 0 \\ \frac{1}{2} & 0 & -\frac{1}{2} & \frac{1}{\sqrt{2}} \\ \frac{1}{2} & 0 & -\frac{1}{2} & -\frac{1}{\sqrt{2}} \end{pmatrix}$$

The compact SVD $A = 2\sqrt{3}\,p_1q_1' + 2\,p_2q_2'$; check:

$$2\sqrt{3}\,p_1q_1' + 2\,p_2q_2' = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}(1,1,1,1) + \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}(1,-1,0,0) = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix} + \begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = A$$

$-p_2$ is also a unit eigenvector of AA' belonging to 4 (Theorem 3(b)), but $2\sqrt{3}\,p_1q_1' - 2\,p_2q_2' \neq A$: P_1 must be computed from Q_1 as AQ_1D^{-1}.

2. Low-rank approximation and principal components

Theorem 5 (Ky Fan)**.** Let $A \in \mathbb{R}^{n\times n}$ be symmetric, $q_1, \ldots, q_n$ orthonormal, $Aq_i = \lambda_i q_i$, $\lambda_1 \geqslant \cdots \geqslant \lambda_n$ (as in Lecture 9, Theorem 13); $1 \leqslant k \leqslant n$, $G \in \mathbb{R}^{n\times k}$, $G'G = I_k$. Then $\mathrm{tr}(G'AG) \leqslant \lambda_1 + \cdots + \lambda_k$, with equality for $G = [q_1 \ \cdots \ q_k]$.

Proof.

$Q = [q_1 \ \cdots \ q_n]$, $Q'AQ = \Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$, $QQ' = I_n$ (Lecture 9, Theorem 13; Lecture 6, Theorem 9); $R = Q'G = (r_{ij}) \in \mathbb{R}^{n\times k}$:

$$\begin{aligned} G = QQ'G = QR, \qquad & R'R = G'QQ'G = G'G = I_k \\ \mathrm{tr}(G'AG) &= \mathrm{tr}(R'Q'AQR) = \mathrm{tr}(R'\Lambda R) \\ &= \textstyle\sum_{j=1}^k \mathrm{col}_j(R)'\Lambda\,\mathrm{col}_j(R) = \sum_{j=1}^k \sum_{i=1}^n \lambda_i r_{ij}^2 && \text{(Lecture 1, Theorem 4(a),(d))} \\ &= \textstyle\sum_{i=1}^n w_i\lambda_i, \qquad w_i = \sum_{j=1}^k r_{ij}^2 \end{aligned}$$

$0 \leqslant w_i \leqslant 1$, $\sum_i w_i = k$:

$$\begin{aligned} & w_i \geqslant 0, \qquad \textstyle\sum_i w_i = \sum_j \sum_i r_{ij}^2 = \mathrm{tr}(R'R) = \mathrm{tr}\, I_k = k && \text{(Lecture 1, Theorem 10(c))} \\ & w_i = (RR')_{ii} && \text{(Lecture 1, Theorem 4(d))} \\ & k = n\colon\ RR' = I_n \implies w_i = 1 && \text{(Lecture 6, Theorem 9)} \\ & k < n\colon\ [R\ T] \text{ is orthogonal, } RR' + TT' = I_n && \text{(Lecture 6, Corollary 11(c))} \\ & \quad \implies w_i + \textstyle\sum_l t_{il}^2 = 1 \implies w_i \leqslant 1 \end{aligned}$$

The bound ($\lambda_1 \geqslant \cdots \geqslant \lambda_n$, $0 \leqslant w_i \leqslant 1$; the right side expands to $\sum_{i\leqslant k}\lambda_i - \sum_i w_i\lambda_i - \lambda_k(k - \sum_i w_i)$); equality:

$$\begin{aligned} \textstyle\sum_{i\leqslant k}\lambda_i - \sum_{i=1}^n w_i\lambda_i &= \textstyle\sum_{i\leqslant k}(\lambda_i - \lambda_k)(1 - w_i) + \sum_{i>k}(\lambda_k - \lambda_i)w_i && (\textstyle\sum_i w_i = k) \\ &\geqslant 0 && \text{(every term is } \geqslant 0) \end{aligned}$$

$$G = [q_1 \ \cdots \ q_k]\colon\ AG = [\lambda_1 q_1 \ \cdots \ \lambda_k q_k] = G\,\mathrm{diag}(\lambda_1, \ldots, \lambda_k)$$
$$\implies G'AG = \mathrm{diag}(\lambda_1, \ldots, \lambda_k),\ \mathrm{tr}(G'AG) = \lambda_1 + \cdots + \lambda_k$$

□

Theorem 6 (Eckart–Young)**.** Let $A \in \mathbb{R}^{m\times n}$, $\operatorname{rank} A = r$, have an SVD $A = P\operatorname{diag}(D,0)Q' = \sigma_1 p_1 q_1' + \cdots + \sigma_r p_r q_r'$ (Theorem 1); let $1 \leqslant k < r$ and $A_k = \sigma_1 p_1 q_1' + \cdots + \sigma_k p_k q_k'$. Then $\operatorname{rank} A_k = k$, and every $B \in \mathbb{R}^{m\times n}$ with $\operatorname{rank} B \leqslant k$ has

$$\|A - B\|_F^2 \geqslant \sigma_{k+1}^2 + \cdots + \sigma_r^2$$

with equality for $B = A_k$.

Proof.

$B = UL$: $s = \operatorname{rank} B \leqslant k < r \leqslant m$ (Lecture 4, Theorem 12(a)); $\mathcal{C}(B) \perp \mathcal{C}(B)^\perp$:

$u_1, \ldots, u_s$: an orthonormal basis of $\mathcal{C}(B)$ (none if $s = 0$)	(Lecture 6, Corollary 11(a))
$\dim \mathcal{C}(B)^\perp = m - s \geqslant k - s$	(Lecture 6, Theorem 5(c))
$u_{s+1}, \ldots, u_k$: $k - s$ vectors of an orthonormal basis of $\mathcal{C}(B)^\perp$	(Lecture 6, Corollary 11(a))
$U = [u_1 \ \cdots \ u_k]$: $U'U = I_k$, $\operatorname{col}_j(B) = U\ell_j$ $(\ell_j \in \mathbb{R}^k)$	
$B = UL$, $L = [\ell_1 \ \cdots \ \ell_n]$	(Lecture 1, Theorem 4(a),(b))

So it suffices: $\|UL - A\|_F^2 \geqslant \sigma_{k+1}^2 + \cdots + \sigma_r^2$ for all $U \in \mathbb{R}^{m\times k}$ with $U'U = I_k$ and all $L \in \mathbb{R}^{k\times n}$.

$U'U = I_k$; Lecture 1, Theorem 10(b),(c):

$$(UL - A)'(UL - A) = L'L - A'UL - L'U'A + A'A = (L - U'A)'(L - U'A) - A'UU'A + A'A$$

$$\|UL - A\|_F^2 = \operatorname{tr}\Big[(L - U'A)'(L - U'A)\Big] - \operatorname{tr}(A'UU'A) + \operatorname{tr}(A'A) \geqslant \operatorname{tr}(A'A) - \operatorname{tr}(U'AA'U)$$

$Q'Q = I_n$, $P'P = I_m$; Lecture 5, Theorem 2; Theorem 5 for AA' and $G = U$:

$$AA' = P\operatorname{diag}(D,0)Q'Q\operatorname{diag}(D,0)'P' = P\operatorname{diag}(\sigma_1^2, \ldots, \sigma_r^2, 0, \ldots, 0)P'$$

$\operatorname{tr}(A'A) = \operatorname{tr}(AA') = \sigma_1^2 + \cdots + \sigma_r^2$	(Lecture 1, Theorem 10(b))
$AA'p_i = \sigma_i^2 p_i$ $(i \leqslant r)$, $AA'p_i = 0$ $(i > r)$	(Lecture 1, Theorem 4(a))
$\sigma_1^2 \geqslant \cdots \geqslant \sigma_r^2 \geqslant 0 = \cdots = 0$: the eigenvalues of AA'	(Lecture 8, Theorem 9(b))
$\operatorname{tr}(U'AA'U) \leqslant \sigma_1^2 + \cdots + \sigma_k^2$	
$\|UL - A\|_F^2 \geqslant \sigma_{k+1}^2 + \cdots + \sigma_r^2$	

Equality, $\operatorname{rank} A_k$: E, $E_k \in \mathbb{R}^{m\times n}$ have $\sigma_{k+1}, \ldots, \sigma_r$ in positions $(k+1, k+1), \ldots, (r,r)$, resp. $\sigma_1, \ldots, \sigma_k$ in $(1,1), \ldots, (k,k)$, zeros elsewhere; the nonzero columns $\sigma_i e_i$ of E_k are independent (Lecture 6, Theorem 7(b)):

$A - A_k = \sum_{i=k+1}^r \sigma_i p_i q_i' = PEQ'$, $A_k = PE_kQ'$	(Lecture 1, Theorem 4(d))
$\|A - A_k\|_F^2 = \operatorname{tr}(QE'P'PEQ') = \operatorname{tr}(E'EQ'Q)$	(Lecture 1, Theorem 10(b))
$= \operatorname{tr}(E'E) = \sigma_{k+1}^2 + \cdots + \sigma_r^2$	(Lecture 1, Theorem 10(c))
$\operatorname{rank} A_k = \operatorname{rank} E_k = k$	(Lecture 4, Theorem 15(d))

□

Likewise $\|A_k\|_F^2 = \operatorname{tr}(E_k'E_k) = \sigma_1^2 + \cdots + \sigma_k^2$: A_k reproduces the proportion $(\sigma_1^2 + \cdots + \sigma_k^2)/(\sigma_1^2 + \cdots + \sigma_r^2)$ of the sum of squares $\|A\|_F^2 = \sum_{i,j} a_{ij}^2$ of the entries of A (Theorem 3(d)), and no B with $\operatorname{rank} B \leqslant k$ loses less than $\sigma_{k+1}^2 + \cdots + \sigma_r^2$.

Proposition 7. Let $X \in \mathbb{R}^{n\times p}$ be a data matrix, $r = \operatorname{rank}(C_nX) \geqslant 1$.

(a) If $C_nX = P\operatorname{diag}(D,0)Q'$ is an SVD, with the compact SVD $\sigma_1p_1q_1' + \cdots + \sigma_rp_rq_r'$, then $(n-1)S = Q\operatorname{diag}(\sigma_1^2,\ldots,\sigma_r^2,0,\ldots,0)Q'$; $\lambda_m = \sigma_m^2/(n-1)$ $(m \leqslant r)$, $\lambda_m = 0$ $(m > r)$; $V = Q$ can be taken in Lecture 9, Theorem 16, and then $z_m = \sigma_m p_m$ $(m \leqslant r)$; $\mathrm{PVE}_m = \sigma_m^2/(\sigma_1^2 + \cdots + \sigma_r^2)$ $(m \leqslant r)$, $\mathrm{PVE}_m = 0$ $(m > r)$.

(b) Conversely, for v_m, λ_m, z_m as in Lecture 9, Theorem 16, let $\sigma_m = \|z_m\| = \sqrt{(n-1)\lambda_m}$ and $p_m = z_m/\sigma_m$ $(m \leqslant r)$. Then $p_1,\ldots,p_r$ extend to an orthogonal P, $C_nX = P\operatorname{diag}(D,0)V'$ $(D = \operatorname{diag}(\sigma_1,\ldots,\sigma_r))$ is an SVD, and $C_nX = \sigma_1p_1v_1' + \cdots + \sigma_rp_rv_r'$ is its compact SVD.

Proof.

(a) $P'P = I_n$; Lecture 9, Proposition 10(a); Lecture 5, Theorem 2:

$$\begin{aligned}
(n-1)S &= (C_nX)'(C_nX) = Q\operatorname{diag}(D,0)'P'P\operatorname{diag}(D,0)Q' \\
&= Q\operatorname{diag}(\sigma_1^2,\ldots,\sigma_r^2,0,\ldots,0)Q' \\
Sq_m &= \tfrac{\sigma_m^2}{n-1}q_m \ (m \leqslant r), \quad Sq_m = 0 \ (m > r) && \text{(Lecture 1, Theorem 4(a))} \\
&\implies \tfrac{\sigma_1^2}{n-1} \geqslant \cdots \geqslant \tfrac{\sigma_r^2}{n-1} > 0 = \cdots = 0 \text{ are } \lambda_1,\ldots,\lambda_p;\ S \neq 0 && \text{(Lecture 8, Theorem 9(b))} \\
&\implies V = Q \text{ is as in Lecture 9, Theorem 16} \\
z_m &= C_nXq_m = \textstyle\sum_i \sigma_i p_i (q_i'q_m) = \sigma_m p_m \quad (m \leqslant r) \\
\mathrm{PVE}_m &= \lambda_m/\textstyle\sum_l \lambda_l = \sigma_m^2/\sum_{l\leqslant r}\sigma_l^2 \ (m \leqslant r),\ 0 \ (m > r) && \text{(Lecture 9, Proposition 17)}
\end{aligned}$$

(b) Lecture 1, Theorems 4(a),(d), 10(c); Lecture 6, Theorem 9; Lecture 9, Proposition 10(a), Theorem 16(a),(d):

$$\begin{aligned}
&r \geqslant 1 \implies C_nX \neq 0 \implies \operatorname{tr}\Big((n-1)S\Big) = \operatorname{tr}\Big((C_nX)'(C_nX)\Big) > 0 \implies S \neq 0 \\
&C_nX = (C_nXV)V' = z_1v_1' + \cdots + z_pv_p' \quad (VV' = I_p) \\
&\|z_m\|^2 = (n-1)\lambda_m\colon\ z_m = 0 \ (m > r); \quad z_m = \sigma_m p_m,\ \sigma_m > 0 \ (m \leqslant r) \\
&\quad (C_nX)'(C_nX)v_m = (n-1)Sv_m = (n-1)\lambda_m v_m \ (SV = V\Lambda), \quad (n-1)\lambda_1 \geqslant \cdots \geqslant (n-1)\lambda_p \\
&\quad \implies V \text{ is as in Lecture 9, Theorem 13, for } (C_nX)'(C_nX), \text{ with the eigenvalues } (n-1)\lambda_m \\
&\quad \implies \sigma_m^2 = (n-1)\lambda_m = \lambda_m\Big((C_nX)'(C_nX)\Big) \ (m \leqslant r) \\
&\quad \implies V,\ D = \operatorname{diag}(\sigma_1,\ldots,\sigma_r) \text{ are as in Theorem 1(b) for } C_nX \\
&P_1 = C_nX[v_1\ \cdots\ v_r]D^{-1} = [z_1/\sigma_1\ \cdots\ z_r/\sigma_r] = [p_1\ \cdots\ p_r]
\end{aligned}$$

Theorem 1(b) extends P_1 to an orthogonal P and gives the two forms.

□

For $1 \leqslant k < r$ and $v_m = q_m$, the A_k of Theorem 6 for $A = C_nX$ is $\sigma_1p_1q_1' + \cdots + \sigma_kp_kq_k' = z_1v_1' + \cdots + z_kv_k'$, with $\sigma_{k+1}^2 + \cdots + \sigma_r^2 = (n-1)(\lambda_{k+1} + \cdots + \lambda_p)$: the scores and loading vectors of the first k principal components give a matrix of rank at most k closest to C_nX.

Example 8. Write the compact SVD of C_5X for the data of Lecture 9, Examples 11 and 18.

Solution.

Proposition 7(b), $n-1 = 4$; $\lambda_m = 25, 9, 5, 1$, $v_m = \frac{1}{2}h_m$, z_m (Lecture 9, Example 18); $r = 4$ ($\lambda_4 = 1 > 0$; Lecture 9, Theorem 16(a)):

$$\begin{aligned}
&\sigma_1 = \sqrt{4\cdot 25} = 10, \quad \sigma_2 = \sqrt{4\cdot 9} = 6, \quad \sigma_3 = \sqrt{4\cdot 5} = 2\sqrt{5}, \quad \sigma_4 = \sqrt{4\cdot 1} = 2 \\
&C_5X = \sigma_1p_1v_1' + \sigma_2p_2v_2' + \sigma_3p_3v_3' + \sigma_4p_4v_4', \qquad \sigma_1p_1v_1' = z_1v_1' = \tfrac{1}{2}(0,5,5,-5,-5)'(1,1,1,1) \\
&\text{check (Theorem 3(d))}\colon\ \sigma_1^2 + \sigma_2^2 + \sigma_3^2 + \sigma_4^2 = 100 + 36 + 20 + 4 = 160 = \|C_5X\|_F^2 = 4\operatorname{tr}S = 4\cdot 40
\end{aligned}$$

with $p_m = z_m/\sigma_m$:

$$p_1 = \frac{1}{2}\begin{pmatrix}0\\1\\1\\-1\\-1\end{pmatrix}, \qquad p_2 = \frac{1}{2}\begin{pmatrix}0\\1\\-1\\1\\-1\end{pmatrix}, \qquad p_3 = \frac{1}{2\sqrt{5}}\begin{pmatrix}4\\-1\\-1\\-1\\-1\end{pmatrix}, \qquad p_4 = \frac{1}{2}\begin{pmatrix}0\\1\\-1\\-1\\1\end{pmatrix}$$

3. The spectral norm and the condition number

Definition 9. The **spectral norm** of $A \in \mathbb{R}^{m\times n}$ is $\|A\|_2 = \sigma_1$, its largest singular value ($\|0\|_2 = 0$); it is not $\|A\|_F$. The **condition number** of an invertible $A \in \mathbb{R}^{n\times n}$ is $\kappa(A) = \|A\|_2\|A^{-1}\|_2$. For $X \in \mathbb{R}^{n\times p}$ with $\operatorname{rank} X = p < n$ and singular values $\sigma_1 \geqslant \cdots \geqslant \sigma_p > 0$, $\kappa(X) = \sigma_1/\sigma_p$ (for $p = n$, Proposition 11(a) gives the same number).

Proposition 10. Let $A \in \mathbb{R}^{m\times n}$.

(a) $\|A\|_2 = \big(\lambda_1(A'A)\big)^{1/2} = \max_{x\neq 0}\|Ax\|/\|x\|$; so $\|Ax\| \leqslant \|A\|_2\|x\|$ for every $x \in \mathbb{R}^n$, with equality at $x = q_1$ if $A \neq 0$ (q_1 the first column of Q in an SVD).

(b) $\|UAV\|_2 = \|A\|_2 = \|A'\|_2$ for orthogonal $U \in \mathbb{R}^{m\times m}$, $V \in \mathbb{R}^{n\times n}$.

Proof. $A = 0$: every statement reads $0 = 0$ or $0 \leqslant 0$. Let $A \neq 0$, so $\operatorname{rank} A \geqslant 1$, with an SVD (Theorem 1(b)) as in Theorem 3.

(a)

$$\begin{aligned}
\lambda_1(A'A) &= \sigma_1^2 && \text{(Theorem 3(a))}\\
\|Ax\|^2/\|x\|^2 = x'A'Ax/x'x &\leqslant \lambda_1(A'A) = \sigma_1^2 \quad (x \neq 0) && \text{(Lecture 9, Theorem 13(a), for } A'A)\\
\|Aq_1\| &= \|\sigma_1 p_1\| = \sigma_1 = \sigma_1\|q_1\| && \text{(Theorem 3(c))}
\end{aligned}$$

So the maximum is $\sigma_1 = \|A\|_2$, and $\|Ax\| \leqslant \|A\|_2\|x\|$ (also for $x = 0$).

(b) By Theorem 3(e), UAV, A and A' have the same largest singular value. □

Proposition 11. (a) Let $A \in \mathbb{R}^{n\times n}$ be invertible. Then A has n singular values $\sigma_1 \geqslant \cdots \geqslant \sigma_n > 0$, $\|A^{-1}\|_2 = 1/\sigma_n$ and $\kappa(A) = \sigma_1/\sigma_n \geqslant 1$. If A is PD, with eigenvalues $\lambda_1 \geqslant \cdots \geqslant \lambda_n > 0$, then $\kappa(A) = \lambda_1/\lambda_n$.

(b) If $X \in \mathbb{R}^{n\times p}$, $\operatorname{rank} X = p$, then $X'X$ is PD and $\kappa(X'X) = \kappa(X)^2$.

Proof.

(a) Lecture 4, Theorem 12(d); Theorem 1(b) with $r = n$; Lecture 2, Theorem 11(b); Lecture 6, Theorem 9:

$$\begin{aligned}
&\operatorname{rank} A = n \implies A = PDQ',\ D = \operatorname{diag}(\sigma_1,\ldots,\sigma_n),\ \sigma_1 \geqslant \cdots \geqslant \sigma_n > 0\\
&A^{-1} = (Q')^{-1}D^{-1}P^{-1} = QD^{-1}P' \implies \|A^{-1}\|_2 = \|D^{-1}\|_2 && \text{(Proposition 10(b))}\\
&\|D^{-1}x\|^2 = \textstyle\sum_i x_i^2/\sigma_i^2 \leqslant \|x\|^2/\sigma_n^2,\ \text{equality at } x = e_n\\
&\quad\implies \|D^{-1}\|_2 = 1/\sigma_n && \text{(Proposition 10(a))}\\
&\kappa(A) = \sigma_1 \cdot \tfrac{1}{\sigma_n} \geqslant 1; \quad A \text{ PD} \implies \sigma_i = \lambda_i,\ \kappa(A) = \lambda_1/\lambda_n && \text{(Theorem 3(e))}
\end{aligned}$$

(b) $\sigma_1 \geqslant \cdots \geqslant \sigma_p > 0$: the singular values of X ($r = p$):

$$\begin{aligned}
&X'X \text{ is PD} && \text{(Lecture 8, Theorem 15(b))}\\
&\text{its eigenvalues are } \sigma_1^2 \geqslant \cdots \geqslant \sigma_p^2 > 0 && \text{(Theorem 3(a))}\\
&\kappa(X'X) = \sigma_1^2/\sigma_p^2 = \kappa(X)^2 && \text{((a); Definition 9)}
\end{aligned}$$

□

Theorem 12. Let $A \in \mathbb{R}^{n\times n}$ be invertible, $Ax = b \neq 0$ and $A(x+\delta x) = b + \delta b$. Then

$$\frac{\|\delta x\|}{\|x\|} \leqslant \kappa(A)\,\frac{\|\delta b\|}{\|b\|}$$

Proof.

$\|My\| \leqslant \|M\|_2\|y\|$ for every M, y (Proposition 10(a)):

$$A\,\delta x = \delta b \implies \delta x = A^{-1}\delta b \implies \|\delta x\| \leqslant \|A^{-1}\|_2\|\delta b\| \qquad \text{(subtract } Ax = b\text{)}$$

$$x \neq 0\ (b \neq 0), \quad \|b\| = \|Ax\| \leqslant \|A\|_2\|x\| \implies \frac{1}{\|x\|} \leqslant \frac{\|A\|_2}{\|b\|}$$

$$\frac{\|\delta x\|}{\|x\|} \leqslant \|A^{-1}\|_2\|A\|_2\,\frac{\|\delta b\|}{\|b\|} = \kappa(A)\,\frac{\|\delta b\|}{\|b\|} \qquad \text{(multiply; all terms } \geqslant 0\text{)}$$

□

A large $\kappa(A)$: a small relative change of b may change the solution of $Ax = b$ by a large relative amount. For the normal equations $X'X\beta = X'y$ (Lecture 11, Definition 1) with $\operatorname{rank} X = p$, $\kappa(X'X) = \kappa(X)^2$ (Proposition 11(b)): forming $X'X$ squares the condition number of X.

Example 13. Let

$$A = \begin{pmatrix} 2 & 0 & 1 & -1 \\ 0 & 2 & 1 & 1 \\ 1 & 1 & 3 & 0 \\ -1 & 1 & 0 & 3 \end{pmatrix}$$

Find $\kappa(A)$, and compare the two sides of Theorem 12 for $x = (1,1,2,0)'$ and $\delta b = (1,1,-1,0)'$.

Solution.

Four products; the four vectors are pairwise orthogonal, of lengths $\sqrt{6}$, $\sqrt{6}$, $\sqrt{3}$, $\sqrt{3}$:

$$A(1,1,2,0)' = 4(1,1,2,0)', \qquad A(-1,1,0,2)' = 4(-1,1,0,2)'$$

$$A(1,1,-1,0)' = (1,1,-1,0)', \qquad A(1,-1,0,1)' = (1,-1,0,1)'$$

divided by their lengths: orthonormal $\implies$ eigenvalues $4,4,1,1$ (Lecture 8, Theorem 9(b))

all eigenvalues $> 0 \implies A$ is PD (Lecture 8, Theorem 14(a))

$\operatorname{rank} A = \#\{i : \lambda_i \neq 0\} = 4$ (Lecture 8, Theorem 9(b))

$\implies A$ is invertible (Lecture 4, Theorem 12(d))

$\kappa(A) = 4/1 = 4$ (Proposition 11(a))

$b = Ax$, $\delta x = A^{-1}\delta b$:

$$b = Ax = 4x = (4,4,8,0)', \qquad A\,\delta b = \delta b \implies \delta x = A^{-1}\delta b = \delta b = (1,1,-1,0)'$$

$$\frac{\|\delta x\|}{\|x\|} = \frac{\sqrt{3}}{\sqrt{6}}, \qquad \frac{\|\delta b\|}{\|b\|} = \frac{\sqrt{3}}{\sqrt{96}} = \frac{1}{4}\cdot\frac{\sqrt{3}}{\sqrt{6}} \implies \frac{\|\delta x\|}{\|x\|} = \kappa(A)\,\frac{\|\delta b\|}{\|b\|}$$

Equality in Theorem 12: the relative change of b is magnified four times in the solution.

Practice. The Exercises and Problems of the textbook, Chapter 15: the singular value decomposition, low-rank approximation, principal components through the SVD, the spectral norm and the condition number.